



College: Health, Science, and Technology

Department: Mathematics, Actuarial Science, and Statistics

Course Type: Face-to-face, Warrensburg Campus

Course Prefix/Course Number/Course Name: ACST 5351 Principles of Data Analysis

Semester and Year: Spring 2026

Number of Credit Hours¹: Three Credit Hours

Catalog Course Description, Prerequisites, Co-requisites, and Other Restrictions: This course is an introduction to data mining. It introduces fundamental strategies and methodologies for data mining along with the concepts underlying them and provides hands-on experience with a variety of descriptive and predictive modeling. Most important data mining techniques such as generalized linear models, decision trees, unsupervised learning, predictive modeling for classification and regression will be covered. The course covers most materials of Exam PA and some topics of Exam SRM from the Societies of Actuaries. This course is co-listed with ACST 4351. Prerequisite(s): ACST 5321 with a grade of C or better.

Instructor Information

Instructor: Yingzhou He, Ph.D., ASA (Pronouns: he/his/him)

Office Phone: (660)-543-4337

Faculty Availability (Face-to-face and Online): Face to Face

Office hours: MTW: 3:30 - 4:30 pm, R: 11:00 am - 12:00 pm, 3:30 - 4:30 pm. **WCM 125.**

University Email Address: yhe@ucmo.edu

Anticipated Response Time for Emails, Assignment Grading, and Feedback:

Task	Expected Response Time
Email Response	Within 24 hours
Grading – Assignments	Within one week
Grading – Tests	Within one week
Feedback with Grades	Within one week

¹The amount of student engagement in coursework to achieve the intended learning outcomes should be appropriate to the credit hours awarded. A **credit hour** is an amount of work represented in intended learning outcomes and verified by evidence of student achievement that is an institutionally-established equivalency that reasonably approximates not less than: (1) one hour of classroom or direct faculty instruction and a minimum of two hours of out-of-class student work each week for approximately fifteen weeks for one semester hour of credit or the equivalent amount of work over a different amount of time; or (2) at least an equivalent amount of work as required in (1) above of this definition for other activities as established by an institution, including laboratory work, internships, practica, studio work, and other academic work leading toward the award of credit hours; or (3) institutionally established reasonable equivalencies for the amount of work described above in paragraph (1) of this definition for the credit hours awarded, as represented by verifiable student achievement of intended learning outcomes.

Course Information & Requirements

Materials

- **Textbook:** An Introduction to Statistical Learning with Applications in R, by G. James, D. Witten, T. Hastie and R. Tibshirani, Springer, 2nd edition 2021.
You can buy or rent the textbook here!
- **Course Slides:** Will be posted on Brightspace.

Student Learning Outcomes (SLO)

(*indicates addressed in this course)

1. Communicate actuarial/statistical ideas clearly and coherently.
2. *Demonstrate the knowledge of the background and principle of solving problems in actuarial/statistical fields.
3. Use actuarial/statistical software packages to solve real-world problems.

Course Objectives

Upon completion of this course, the student will

1. Understand principles of data mining.
2. Understand data mining techniques such as generalized linear models, decision trees, unsupervised learning, and predictive modeling for classification and regression.
3. Formulate and solve problems that involve data mining techniques.
4. Identify data mining models in diverse fields of applications.
5. Understand and be capable of interpreting codes and outputs from data mining methods in R.
6. Apply data mining methods in R to some real-world predictive problems.

Instructional Methods / Activities / Assessments

1. **Assignments (35%):** Students are supposed to access and submit the assignments on BrightSpace. Each assignment consists of theoretical and coding questions. The assignment will be graded based on both the completeness and correctness. Students are highly recommended to finish their assignments using **RMarkdown**. Late submissions will be given only in the case of an unavoidable emergency.
2. **Midterm (15%) & Final (20%) :** A **midterm test** and a **comprehensive final exam** are scheduled for this class. Students can bring a **two-sided handwritten A4 sheet** for the midterm and the final exam. The midterm will be taken during the class and last **50 minutes**. The final exam will be held in the final week and will last 120 minutes. Make-up tests will be given only in the case of an unavoidable emergency to students who regularly attend class. **The graduate students are supposed to solve additional challenging problems in both exams.**

3. **Individual Project (30%)**: The project is for the entire semester. Students will work on the project **Titanic - Machine Learning from Disaster**, a classic classification data mining problem from **Kaggle**. The individual project will be graded based on the following:

- **Project Presentation 20%**: The project presentation will be held the last week before the final. Each student must prepare and make a 15 to 20-minute presentation.
- **Beat the Benchmark 40%**: The project uses an accuracy metric. Students must obtain a model that beats the benchmark **0.80**. Students can submit the prediction result to test the model anytime on the Kaggle project page, which can help the students select the best and final model.
- **Project Report 40%**: Students must submit a project report by the end of the semester in **PDF** format, along with your model in R or RMarkdown (recommended). Students must include tables, graphs, and model outputs in the report. Students may refer to the models and reports generated by others for this contest posted on the Kaggle.

More details and instructions for the individual project will be released by the instructor throughout the semester.

Grading Practices

- Your grade will be based on the following factors:

Assignments	35%
Midterm	15%
Individual Project	30%
Final Exam	20%

- The grading scale will be:
A: 90 – 100% B: 80 – 90% C: 70 – 80% D: 60 – 70% F: 0 – 60%

Artificial Intelligence Statement

Understanding how and when to use generative AI tools (such as ChatGPT, Gemini, etc.) to brainstorm, explore responses, and find information is quickly emerging as an important skill for future professions. To that end, the use of AI tools is permitted only for personal study support, such as reviewing concepts, providing more examples and explanations or clarifying steps from the course materials. However, you must independently complete all graded assessments without assistance during submission. Do not use AI to submit answers directly on any graded assessment. AI-generated submissions are not permitted and will be treated as academic misconduct. If you have any questions or need clarification on what constitutes permissible use of AI tools, please consult with the instructor before proceeding with any coursework.

Course Attendance Policy

Students are expected to participate fully in class and to help create an atmosphere of learning for everyone. Students are further expected to treat everyone within the classroom with

respect and dignity, discuss controversial issues or views without insulting the integrity of another, and to attend to whatever task is at hand within the classroom. Those guidelines also apply to the instructor. Behaviors which detract from the academic atmosphere can result in you being asked to leave a particular class period, could result in your removal from the course, or a reduction in the final grade. Staying active and getting involved in the class would be helpful to understand the materials thoroughly. Please do not hesitate to ask questions in class or stop by my office for more support.

Course Late Work Policy

Late submissions will be given only in the case of an unavoidable emergency. Students must notify the instructor ahead of time.

Outline of Course Content

Chapter	Content	Sections
2	Statistical Learning	Sections 1-3
3	Linear Regression	Sections 1-6
4	Classification	Sections 1-3 and parts of 6
5	Resampling Methods	Sections 1 and 3 (excluding 5.3.4)
6	Linear Model Selection and Regularization	Sections 1-5
8	Tree Based Methods	Sections 1-3 (excluding 8.2.4, 8.2.5, and 8.3.5)
12	Unsupervised Learning	Sections 1-2, 4-5 (excluding 12.5.2)

University Policies

ADA

Students with documented disabilities who are seeking accommodations should contact the Office of Accessibility Services at Union 224, 660-543-4983 or access@ucmo.edu. If you want to share information about your needs that I should be aware of, such as emergency medical information or special arrangements for field trips or internships, please see me privately after class or during office hours. Please contact Accessibility Services, 660-543-4983 or access@ucmo.edu, at least 72 hours in advance if you will require accommodations due to disability.

Title IX

The University of Central Missouri seeks to foster a safe and healthy environment built on mutual respect and trust. Sex discrimination, including sexual harassment, sexual violence and other forms of sexual misconduct will not be tolerated. All faculty and most staff are considered mandated reporters by the University and must disclose all information they receive about sexual misconduct to the Title IX Coordinator. **As a faculty or staff member of the University, I am a mandated reporter. This means I am required to report information shared with me regarding sex discrimination and sexual misconduct.**

If you, or someone you know, has experienced sex discrimination or sexual misconduct, please know assistance and options are available. UCM strongly encourages all members of the

community to seek support and report incidents of this nature to the Title IX Coordinator. Anyone who wishes to report sexual misconduct, to learn more about the University process and options available, or to utilize a confidential resource, please visit [.ucmo.edu/titleix](https://ucmo.edu/titleix).

If you are pregnant, you are also protected under Title IX. You should contact the Office of Accessibility Services at Union 224, 660-543-4983 or access@ucmo.edu for accommodations related to your pregnancy.

Access, Opportunity, and Community

The University of Central Missouri's administrators, faculty, and staff desire to have a campus environment that recognizes all dimensions of diversity and inclusiveness. All students are welcome in our classes, and differences are acknowledged rather than erased or denied. Dimensions of diversity can include sex, race, age, national origin, ethnicity, gender identity and expression, intellectual and physical ability, sexuality, income, faith and non-faith perspectives, socio-economic class, primary language, family status, military experience, and more. Inclusive learning is facilitated by creative and innovative thought and mutual respect; being in this class means that you, your faculty member, and your peers strive to foster a welcoming and equitable environment for all.

Academic Honesty

This instructor will enforce UCM's Academic Honesty Policy as described in the Academic Policy Library and UCM Student Handbook.

Student Conduct

Students must abide by University rules, local, state, and federal laws. A full description of student rights and responsibilities is available in the Student Handbook.